Hair2Harvest – Revised System Design

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Summary of changes

Suppliers become records that staff manage, not users who log in, because the interviewed shops are busy, mall-based or short-staffed and do not want extra tasks. Every other change below follows from that: Hair2Harvest comes to the shop, does the work there, and tells the shop what happened.

#	Change	Interview evidence	Optimal approach
1	Suppliers have no accounts	Shops have work shifts and cannot go to the barangay (Q13); "they don't want to be bothered"	Staff register the shop on site in one short form (about 2 minutes); no password, no supplier login
2	Pickup schedule is set per shop	Pickup needs range from nightly (mall shop) to twice weekly to weekly (Q9, Q13)	Each shop gets its own pickup rule: days, time window, access notes; default twice a week
3	Collector goes to the shop	Shops will not travel to the barangay (Q13)	Remove drop-off flow; add a collector's route list on a phone, with offline support
4	Pickups record sacks and kilograms	Volume is described only as sacks per day or week (Q2); real mass unknown (Part E)	Record sack count and weighed kg for every pickup; subtract sack weight
5	Mixed waste is checked and graded	Hair mixed with tissue and plastic in shops and in photos F-01 and F-04 (Q4)	Give each shop a labeled hair-only sack; collector grades contamination at pickup
6	Treated hair is flagged	About 15% of clients have treated hair (Q11)	Add a treated/untreated flag; only untreated hair enters batches by default

#	Change	Interview evidence	Optimal approach
7	Shops are contacted without logins	Shops want a fixed schedule and a pickup that does not depend on them (Q13)	Staff send templated SMS or Messenger reminders; shop confirms with a tap on the collector's phone
8	Consent is recorded at registration	Photo and participation permission is still blank (Part E)	Consent checkbox, date and photo permission captured by staff, stored per shop
9	Map becomes a simple shop list first	Many barbershops, only 9 candidates identified, 6 interviewed	Candidate-to-enrolled status list; static pins stay optional
10	Roles reduced from five to four	Supplier no longer logs in	Admin, Staff, Operator, Researcher; shops get an optional read-only pickup link instead of a role

Change 1: Suppliers have no accounts

Barbershops are registered by Hair2Harvest staff and never log in. Registration, pickup history and status all live in records that staff and admins manage.

What changes in the original design

- Module 1 (Auth and Access): remove the Supplier role and any supplier sign-up, login or password reset.
- Module 2 (Supplier Registration): replace the multi-step self-registration and supplier dashboard with a one-screen, staff-filled form.
- Remove "estimated weekly hair volume" from registration. Shops can only describe volume as sacks, so the system measures it from the first pickups instead.
- Add a status for each shop: Candidate, Enrolled, Paused, Withdrawn. The 9 candidate shops already identified can be loaded as Candidates before the first visit.

Optimal approach

1. Pre-load candidate shops, then staff visit and finish registration on a phone. The form asks only for shop name, contact person, mobile number, address, opening hours, pickup rule (Change 2) and consent. Target: under 2 minutes, with no typing for fields that can be picked from a list.

2. The system generates a shop code (for example HB-001) and the staff member prints or writes it on the shop's hair sack. The code is the shop's identity, so no username or password is needed.
3. Admin reviews and approves the registration afterwards, so the approval step stays but moves from the shop to the team.
4. Record consent at registration: agreed to separate and supply hair, date, who explained it, and whether photos of the shop's waste container may be used in the thesis. This fills the consent gaps listed in Part E of the evidence file.
5. Optional, low priority: a read-only page per shop, opened from a QR code on the sack, showing the last pickups and weights. It needs no login and exposes nothing but that shop's own totals.

Changes 2 and 3: Pickup schedule is set per shop

Pickup frequency differs sharply between shops, so a single schedule will not fit. One mall-based shop said its mall clears waste every day and asked for nightly pickup or its hair will pile up. Others suggested twice a week with a fixed schedule, or weekly. Shops also said the collector must come to them.

What changes in the original design

- Module 2 stops storing "collection preferences" as a loose note. Each shop gets a structured pickup rule.
- Module 3 (Collection and Inventory) generates recurring pickups from those rules, grouped by day, and gives the collector a route list instead of a drop-off process.
- Remove any step that asks the shop to bring hair to the barangay or processing site.

Pickup rule field	Example	Why
Frequency	Twice a week, or nightly	Matches Q9 and Q13 answers
Days	Tue and Fri	Shops asked for a specific schedule
Time window	After closing, or 8 to 9 PM	The mall shop wants nightly pickup because the mall clears waste daily
Access notes	Mall guard, loading dock, who to ask	Shop staff cannot always be present

Pickup rule field	Example	Why
Peak-day flag	Weekend heavy	Weekends and Sundays are busiest (Q1)

Optimal approach

1. Set the default to twice a week for ordinary shops and use the shop's own rule when it differs. Twice a week matches the middle of the range and keeps a sack from sitting longer than a week.
2. Treat the mall shop as a special case. If nightly pickup is too costly for the pilot, give the shop a second sack and a shorter rule (for example, collect on Monday and Thursday) and confirm with the shop that it can hold the sack out of the mall's daily waste stream. Do not assume this works; ask.
3. Build the daily route list by sorting that day's pickups by time window, then by barangay area. This is a plain ordered list, which keeps the "no GIS routing" decision from the original design.
4. Let staff mark a pickup as Skipped (shop closed, no hair yet) or Rescheduled, with a reason. A skipped pickup should not count as a failure by the shop.
5. Keep one visible rule: Hair2Harvest never asks a shop to travel. If a pickup is missed, the collector returns, not the shop.

Changes 4 to 6: What the collector records at the shop

The interviews show three practical facts: hair goes into the same sack or bin as tissue, plastic and packaging; shops describe volume in sacks, not kilograms; and about 15% of clients have treated hair. The pickup record in Module 3 must capture all three at the shop, in seconds.

Field added to each pickup	Type	Why
Sack count	Number	Matches how shops describe volume (Q2)
Gross and net weight (kg)	Numbers	Net = gross minus a standard sack weight; gives the real mass that Part E asks for

Field added to each pickup	Type	Why
Contamination grade	Clean, Minor, Mixed	Photos F-01 and F-04 show mixed waste
Treated hair present	Yes, No, Unsure	About 15% of clients have treated hair (Q11)
Photo	Image	Evidence log with shop code and date, replacing the blank photo log
Shop acknowledgment	Tap or initials on collector's phone	Confirms the pickup without any shop login
Rejected reason	Pick list	Mixed waste, wet, treated, too small

Optimal approach

1. Give each enrolled shop a dedicated, labeled hair sack or bin with its shop code. This is the single best fix for mixed waste, because the interviews show shops separate hair only when a container exists for it (Q3, Q5).
2. Grade contamination at pickup rather than refusing the sack. A Mixed grade triggers a short coaching note, not a penalty, so the shop is not bothered. Only Clean and Minor sacks go to the raw-hair inventory.
3. Keep treated hair out of batches by default. The system flags a sack as Treated or Unsure and holds it in a separate stock line until the specialist says whether it is acceptable (the open question in Part E).
4. Use the phone camera for the photo log. Each photo is stored against the pickup, so the shop code, date, photographer and permission are filled automatically.
5. Make the pickup form work offline. Malls and back rooms have weak signal, so the React app should save pickups on the device and sync when connected.

Change 7: Contacting shops without logins

Module 11 (Notifications) currently sends in-app alerts to every role. Shops have no accounts, so they never see an in-app alert. Shop-facing messages must travel through channels the owner already uses.

Optimal approach

1. Store one mobile number and a preferred channel (SMS, Messenger or Viber, call) per shop at registration.
2. Start with staff-sent messages from templates. The system builds the message ("Pickup tonight 8 to 9 PM, shop HB-001") and opens it in the phone's messaging app with one tap. This costs nothing and needs no SMS gateway.
3. Send only three message types: pickup reminder, pickup missed or rescheduled, and pickup recorded (sacks and kg). Keep each under two lines so shops are not bothered.
4. Let shops reply "skip" or "not full" instead of using a portal. Staff then mark the pickup Skipped.
5. Add an automated SMS gateway only after the pilot proves the volume justifies the cost.
6. Address the shop's one stated concern (will the hair really become fertilizer, Q10) with a short printed or messaged update after each batch result, not with a dashboard. Admin approval of lab results already exists, so the update can reuse that status.

Changes 9 and 10: Map and roles

Map view (Module 4). The group has a list of 9 candidate shops, six were interviewed, and nobody has confirmed barangay boundaries. A map adds little until the list is verified.

- Optimal approach: build a shop list with status (Candidate, Enrolled, Paused, Withdrawn) first and keep the Leaflet map as an optional view of that same list. Pins use the status colors already planned. No routing and no live GPS, as before.
- Add a field for whether a shop is inside the pilot barangay, so the sampling table in the paper uses only eligible shops.

Roles (Module 1). Five roles become four because the Supplier role disappears.

- Admin: approves shops and batches, edits thresholds, reads the audit log.
- Staff: registers shops, schedules and records pickups.
- Operator: runs batches and the machine. One person can hold both Staff and Operator.
- Researcher: enters lab results and manages experiments.
- Optional: a read-only link per shop (no login) for pickup history, as in Change 1. It is a link, not a role.

Records that change. The Supplier record now stores a contact person, mobile number, preferred channel, pickup rule, consent date, photo permission, status and the staff member who registered it. It links to no user account.

Revised modules and access

The design stays at 13 modules and the same stack (MongoDB, Express.js, React.js, Node.js). Only the modules below change.

Module	Change
1 Auth and Access	Supplier role and supplier login removed; four roles remain
2 Supplier (Barbershop) Registration	Staff-filled one-screen form; shop code; status; consent and photo permission; structured pickup rule; no supplier dashboard
3 Collection and Inventory	Per-shop recurring pickups; collector route list; sack count, net kg, contamination grade, treated flag, photo, shop acknowledgment; offline entry; separate stock line for treated hair
4 Map View	Optional view of the shop list; pilot-barangay eligibility field
11 Notifications	In-app alerts for staff only; shop messages are templated SMS or Messenger sent by staff
13 Dashboard	Adds shops by status, pickups missed or skipped, and contamination grade trend
5 to 10, 12	Unchanged, except Batch (5) draws only from Clean and Minor, untreated stock by default

Role access after the changes:

Module	Admin	Staff	Operator	Researcher
Shop registration	Approve, manage	Register, edit	View	View
Collection and inventory	Full	Record pickups	View	View
Batch management	Full	View	Create, update	View

Module	Admin	Staff	Operator	Researcher
Machine control	View	None	Operate	View
Lab results	View	None	None	Enter, upload
Quality approval	Edit rules	None	View status	View
Experiments	View	None	None	Manage
Reports and audit	Full	Limited	Limited	Reports

Open items to verify

- [ ] Confirm which of the 9 candidate shops are inside the pilot barangay and which six were interviewed.
- [ ] Ask the specialist whether treated hair (about 15% of clients) is acceptable in compost, and in what share.
- [ ] Decide how to serve the mall shop that wants nightly pickup, or accept a twice-weekly rule with two sacks.
- [ ] Weigh sacks over several pickups to replace the sack-based estimates with measured kilograms.
- [ ] Get the barangay's written answer on whether a separate hair program exists.
- [ ] Check the screenshot claims about P.D. 856 and DSQC collection against official sources before the paper cites them.
- [ ] Choose the sack weight used to compute net kg.